

Oscar J. Escobar

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SKILLS

- Skills
 - Deep Learning, ML, NLP
 - Flask (Web App)
 - MLOps & LLMOps
 - Bayesian Stats & Data Assimilation & Time Series
 - Modeling with Dynamics & Optimal Control
 - AI: LLMs, MCP, RAG, Tool Calling, Chroma DB
- Coding/Software:
 - Expert: Python, Git/GitHub, Unix Shell/Bash
 - Proficient: Flask, SciKit Learn, PyTorch, StatsModels, Optuna, Pandas, Ollama & OpenAI API; SQL
 - Experience with: AWS, HTML, C++, R, PySpark, Oracle, Excel & Power BI, Tachyon Studio

PROFESSIONAL EXPERIENCE

AI Frontend Engineer & App Developer

June 2025 — Aug 2025

Wells Fargo—intelligent Monitoring & Automation Team

Minneapolis, MN

- Co-authored PY-HTML module that automates reporting for a team 24, improving efficiency by 87%
- Built Flask interactive tool to analyze process data, producing tables and time-series plots of activity for trend analysis
- Tested various LLMs & hyperparameters in Tachyon Studio for agent tool creation

Machine Learning Intern Developer

June 2024 — Aug 2024

Wells Fargo—Advanced Analytics & Solutions Team

Minneapolis, MN

- Crafted NLP text classification model that achieved 77% cross-validated accuracy on high-error-prone data
- Leveraged NLP and statistical analysis to perform feature engineering and presented NLP results and findings to team
- Wrote an executive summary detailing overall team results, findings, and suggestions presented to upper management

Machine Learning & Data Science Student Engineer

April 2023 — April 2024

Family Search—Automated Content Extraction Team

Lehi, UT

- Performed data analysis on ML model output using self-made Python script that compared over 100 records at a time
- Identified model improvement areas that led to an increase in correct predictions of 20-30% using Python script
- Spearheaded efforts and meetings with a team of 3 data labelers to create training data and define project benchmarks

RESEARCH & PROJECTS

AI/LLM/NLP

- Co-crafted agent with Expo (JS) that suggests academic improvement based on grade pattern that won 2nd place
- Investigating the robustness of RLHF, DPO, and IPO on small base LLM using noisy data from Stanford

Data Science & Modeling

- Predicted the location of a submarine on a 2D grid using particle filters
- Predicted weather patterns and surrogate crop yield to high accuracy using time series analysis (ARMA & SARIMAX)

Deep Learning

Dec 2025 — Present

- [Co-published](#) a study of deep reinforcement learning using various decaying greedy policies and replay selection
- Performing a case study of NPG, TRPO, and PPO on robotic walker environment

Research Assistant

Sept 2024 — Present

BYU Mathematics Department

Provo, UT

- Crafting Markov-Chain-Monte-Carlo model to find multivariable roots of nonlinear system
- Developing MLOps architecture to train neural network to solve nonlinear PDE

EDUCATION

Bachelor of Science: Applied and Computational Mathematics

Dec 2025

Brigham Young University

Provo, UT

- 2023-2024 Math Class President; BYU SIAM Chapter Vice-president; AI & Data-Science Club Member
- Minor: ML & AI
- Relevant Courses: MLOps & LLMOps; Agentic AI Applications